SG No. 137 D_{4h}^{15} $P4_2/nmc$ [tetragonal]

* plus set: $+[0, 0, 0]$
* Wyckoff site: `2a`, site symmetry: `-4m2`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `-4m2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[1,2,-7,-8,-11,-12,13,14]`
2	$[0, 0, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[3,4,-5,-6,-9,-10,15,16]`

Table 2: Wyckoff bond: `4b@2a`, bond symmetry: `2mm.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[1,-2,13,-14]`
2	$[0, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[3,-4,-15,16]`
3	$[-X, 0, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[5,-6,9,-10]`
4	$[0, X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[7,-8,-11,12]`

Table 3: Wyckoff bond: `4c@2a`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[1,-2,7,-8]`
2	$[-X, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[3,-4,5,-6]`
3	$[-X, -X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[9,-10,15,-16]`
4	$[X, -X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[11,-12,13,-14]`

Table 4: Wyckoff bond: `8d@2a`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[1,13]`
2	$[-X, 0, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[2,14]`
3	$[0, X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[3,16]`
4	$[0, -X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[4,15]`
5	$[-X, 0, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[5,9]`
6	$[X, 0, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[6,10]`
7	$[0, X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[7,12]`
8	$[0, -X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[8,11]`

Table 5: Wyckoff bond: `8e@2a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[1,-8]`
2	$[-X, -X, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[2,-7]`
3	$[-X, X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[3,-6]`
4	$[X, -X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[4,-5]`

continued ...

Table 5

No.	vector	center	mapping
5	$[-X, -X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[9,-16]`
6	$[X, X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[10,-15]`
7	$[X, -X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[11,-14]`
8	$[-X, X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[12,-13]`

Table 6: Wyckoff bond: `8f@2a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[1,-2]`
2	$[-Y, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[3,-4]`
3	$[-X, Y, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[5,-6]`
4	$[Y, X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[7,-8]`
5	$[-X, -Y, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[9,-10]`
6	$[Y, -X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[11,-12]`
7	$[X, -Y, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[13,-14]`
8	$[-Y, -X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[15,-16]`

Table 7: Wyckoff bond: `16g@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[2]`
3	$[-Y, X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[3]`
4	$[Y, -X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[4]`
5	$[-X, Y, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[5]`
6	$[X, -Y, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[6]`
7	$[Y, X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[7]`
8	$[-Y, -X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[8]`
9	$[-X, -Y, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[9]`
10	$[X, Y, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[10]`
11	$[Y, -X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[11]`
12	$[-Y, X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[12]`
13	$[X, -Y, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[13]`
14	$[-X, Y, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[14]`
15	$[-Y, -X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[15]`
16	$[Y, X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[16]`

* Wyckoff site: `2b`, site symmetry: `-4m2`

Table 8: Wyckoff bond: `2a@2b`, bond symmetry: `-4m2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,2,-7,-8,-11,-12,13,14]`
2	$[0, 0, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[3,4,-5,-6,-9,-10,15,16]`

Table 9: Wyckoff bond: `4b@2b`, bond symmetry: `2mm.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,-2,13,-14]`
2	$[0, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[3,-4,-15,16]`
3	$[-X, 0, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[5,-6,9,-10]`
4	$[0, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[7,-8,-11,12]`

Table 10: Wyckoff bond: `4c@2b`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,-2,7,-8]`
2	$[-X, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[3,-4,5,-6]`
3	$[-X, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[9,-10,15,-16]`
4	$[X, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[11,-12,13,-14]`

Table 11: Wyckoff bond: `8d@2b`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,13]`
2	$[-X, 0, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[2,14]`
3	$[0, X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[3,16]`
4	$[0, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[4,15]`
5	$[-X, 0, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[5,9]`
6	$[X, 0, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[6,10]`
7	$[0, X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[7,12]`
8	$[0, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[8,11]`

Table 12: Wyckoff bond: `8e@2b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,-8]`
2	$[-X, -X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[2,-7]`
3	$[-X, X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[3,-6]`
4	$[X, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[4,-5]`
5	$[-X, -X, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[9,-16]`
6	$[X, X, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[10,-15]`
7	$[X, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[11,-14]`
8	$[-X, X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[12,-13]`

Table 13: Wyckoff bond: `8f@2b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[3,-4]`

continued ...

Table 13

No.	vector	center	mapping
3	$[-X, Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[15,-16]`

Table 14: Wyckoff bond: `16g@2b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[16]`

* Wyckoff site: `4c`, site symmetry: `2mm`.

Table 15: Wyckoff bond: `4a@4c`, bond symmetry: `2mm`.

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,-2,13,-14]`
2	$[0, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[3,-4,-15,16]`
3	$[-X, 0, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[5,-6,9,-10]`
4	$[0, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[7,-8,-11,12]`

Table 16: Wyckoff bond: `4b@4c`, bond symmetry: `2mm`.

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,2,13,14]`
2	$[0, 0, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[3,4,15,16]`
3	$[0, 0, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[5,6,9,10]`
4	$[0, 0, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[7,8,11,12]`

Table 17: Wyckoff bond: `8c@4c`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,13]`
2	$[-X, 0, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[2,14]`
3	$[0, X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[3,16]`
4	$[0, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[4,15]`
5	$[-X, 0, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[5,9]`
6	$[X, 0, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[6,10]`
7	$[0, X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[7,12]`
8	$[0, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[8,11]`

Table 18: Wyckoff bond: `8d@4c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[15,-16]`

Table 19: Wyckoff bond: `16e@4c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[16]`

* Wyckoff site: `4d`, site symmetry: `2mm.`

Table 20: Wyckoff bond: `4a@4d`, bond symmetry: `2mm.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,-2,13,-14]`
2	$[0, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[3,-4,-15,16]`
3	$[-X, 0, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[5,-6,9,-10]`
4	$[0, X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[7,-8,-11,12]`

Table 21: Wyckoff bond: `4b@4d`, bond symmetry: `2mm.`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,2,13,14]`
2	$[0, 0, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[3,4,15,16]`
3	$[0, 0, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[5,6,9,10]`
4	$[0, 0, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[7,8,11,12]`

Table 22: Wyckoff bond: `8c@4d`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,13]`
2	$[-X, 0, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[2,14]`
3	$[0, X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[3,16]`
4	$[0, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[4,15]`
5	$[-X, 0, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[5,9]`
6	$[X, 0, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[6,10]`
7	$[0, X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[7,12]`
8	$[0, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[8,11]`

Table 23: Wyckoff bond: `8d@4d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[15,-16]`

Table 24: Wyckoff bond: `16e@4d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[2]`

continued ...

Table 24

No.	vector	center	mapping
3	$[-Y, X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[3]
4	$[Y, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[4]
5	$[-X, Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	[5]
6	$[X, -Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	[6]
7	$[Y, X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	[7]
8	$[-Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	[8]
9	$[-X, -Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	[9]
10	$[X, Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	[10]
11	$[Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	[11]
12	$[-Y, X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	[12]
13	$[X, -Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	[13]
14	$[-X, Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	[14]
15	$[-Y, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[15]
16	$[Y, X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[16]

* Wyckoff site: 8e, site symmetry: -1

Table 25: Wyckoff bond: 8a@8e, bond symmetry: -1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	[1,-9]
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[2,-10]
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[3,-11]
4	$[Y, -X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[4,-12]
5	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	[5,-13]
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	[6,-14]
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[7,-15]
8	$[-Y, -X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	[8,-16]

* Wyckoff site: 8f, site symmetry: ..2

Table 26: Wyckoff bond: 8a@8f, bond symmetry: ..2

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[x, -x, \frac{1}{4}\right]$	[1,-8]
2	$[-X, -X, Z]$	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{4}\right]$	[2,-7]
3	$[-X, X, Z]$	$\left[x+\frac{1}{2}, x, \frac{3}{4}\right]$	[3,-6]
4	$[X, -X, Z]$	$\left[-x, \frac{1}{2}-x, \frac{3}{4}\right]$	[4,-5]
5	$[-X, -X, -Z]$	$\left[-x, x, \frac{3}{4}\right]$	[9,-16]
6	$[X, X, -Z]$	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{3}{4}\right]$	[10,-15]
7	$[X, -X, -Z]$	$\left[\frac{1}{2}-x, -x, \frac{1}{4}\right]$	[11,-14]
8	$[-X, X, -Z]$	$\left[x, x+\frac{1}{2}, \frac{1}{4}\right]$	[12,-13]

Table 27: Wyckoff bond: `8b@8f`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, 0]$	$\left[x, -x, \frac{1}{4}\right]$	`[1,8]`
2	$[-X, X, 0]$	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{4}\right]$	`[2,7]`
3	$[X, X, 0]$	$\left[x+\frac{1}{2}, x, \frac{3}{4}\right]$	`[3,6]`
4	$[-X, -X, 0]$	$\left[-x, \frac{1}{2}-x, \frac{3}{4}\right]$	`[4,5]`
5	$[-X, X, 0]$	$\left[-x, x, \frac{3}{4}\right]$	`[9,16]`
6	$[X, -X, 0]$	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{3}{4}\right]$	`[10,15]`
7	$[-X, -X, 0]$	$\left[\frac{1}{2}-x, -x, \frac{1}{4}\right]$	`[11,14]`
8	$[X, X, 0]$	$\left[x, x+\frac{1}{2}, \frac{1}{4}\right]$	`[12,13]`

Table 28: Wyckoff bond: `16c@8f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, -x, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{4}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[x+\frac{1}{2}, x, \frac{3}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[-x, \frac{1}{2}-x, \frac{3}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, \frac{1}{2}-x, \frac{3}{4}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x+\frac{1}{2}, x, \frac{3}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{4}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[x, -x, \frac{1}{4}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[-x, x, \frac{3}{4}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{3}{4}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{1}{2}-x, -x, \frac{1}{4}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[x, x+\frac{1}{2}, \frac{1}{4}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x, x+\frac{1}{2}, \frac{1}{4}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{1}{2}-x, -x, \frac{1}{4}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{3}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[-x, x, \frac{3}{4}\right]$	`[16]`

* Wyckoff site: `8g`, site symmetry: `.m.`

Table 29: Wyckoff bond: `8a@8g`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[0, X, Z]$	$\left[\frac{1}{4}, y, z\right]$	`[1,14]`
2	$[0, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{2}-y, z\right]$	`[2,13]`
3	$[-X, 0, Z]$	$\left[\frac{1}{2}-y, \frac{1}{4}, z+\frac{1}{2}\right]$	`[3,15]`
4	$[X, 0, Z]$	$\left[y, \frac{1}{4}, z+\frac{1}{2}\right]$	`[4,16]`
5	$[0, X, -Z]$	$\left[\frac{3}{4}, y+\frac{1}{2}, -z\right]$	`[5,10]`
6	$[0, -X, -Z]$	$\left[\frac{3}{4}, -y, -z\right]$	`[6,9]`
7	$[X, 0, -Z]$	$\left[y+\frac{1}{2}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[7,11]`
8	$[-X, 0, -Z]$	$\left[-y, \frac{3}{4}, \frac{1}{2}-z\right]$	`[8,12]`

Table 30: Wyckoff bond: `8b@8g`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[\frac{1}{4},\ y,\ z\right]$	`[1,-14]`
2	$[-X,\ 0,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ z\right]$	`[2,-13]`
3	$[0,\ X,\ 0]$	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[3,-15]`
4	$[0,\ -X,\ 0]$	$\left[y,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[4,-16]`
5	$[-X,\ 0,\ 0]$	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -z\right]$	`[5,-10]`
6	$[X,\ 0,\ 0]$	$\left[\frac{3}{4},\ -y,\ -z\right]$	`[6,-9]`
7	$[0,\ X,\ 0]$	$\left[y+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[7,-11]`
8	$[0,\ -X,\ 0]$	$\left[-y,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[8,-12]`

Table 31: Wyckoff bond: `16c@8g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{4},\ y,\ z\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ z\right]$	`[2]`
3	$[-Y,\ X,\ Z]$	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[3]`
4	$[Y,\ -X,\ Z]$	$\left[y,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[4]`
5	$[-X,\ Y,\ -Z]$	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -z\right]$	`[5]`
6	$[X,\ -Y,\ -Z]$	$\left[\frac{3}{4},\ -y,\ -z\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[y+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[7]`
8	$[-Y,\ -X,\ -Z]$	$\left[-y,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[8]`
9	$[-X,\ -Y,\ -Z]$	$\left[\frac{3}{4},\ -y,\ -z\right]$	`[9]`
10	$[X,\ Y,\ -Z]$	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -z\right]$	`[10]`
11	$[Y,\ -X,\ -Z]$	$\left[y+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[11]`
12	$[-Y,\ X,\ -Z]$	$\left[-y,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[12]`
13	$[X,\ -Y,\ Z]$	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ z\right]$	`[13]`
14	$[-X,\ Y,\ Z]$	$\left[\frac{1}{4},\ y,\ z\right]$	`[14]`
15	$[-Y,\ -X,\ Z]$	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[15]`
16	$[Y,\ X,\ Z]$	$\left[y,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[16]`

* Wyckoff site: `16h`, site symmetry: `1`

Table 32: Wyckoff bond: `16a@16h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ y,\ z]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	`[2]`
3	$[-Y,\ X,\ Z]$	$\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$	`[3]`
4	$[Y,\ -X,\ Z]$	$\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[4]`
5	$[-X,\ Y,\ -Z]$	$\left[-x,\ y+\frac{1}{2},\ -z\right]$	`[5]`
6	$[X,\ -Y,\ -Z]$	$\left[x+\frac{1}{2},\ -y,\ -z\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[7]`
8	$[-Y,\ -X,\ -Z]$	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	`[8]`
9	$[-X,\ -Y,\ -Z]$	$[-x,\ -y,\ -z]$	`[9]`
10	$[X,\ Y,\ -Z]$	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	`[10]`
11	$[Y,\ -X,\ -Z]$	$\left[y+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	`[11]`

continued ...

Table 32

No.	vector	center	mapping
12	$[-Y,\, X,\, -Z]$	$\left[-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[12]
13	$[X,\, -Y,\, Z]$	$\left[x,\, \frac{1}{2}-y,\, z\right]$	[13]
14	$[-X,\, Y,\, Z]$	$\left[\frac{1}{2}-x,\, y,\, z\right]$	[14]
15	$[-Y,\, -X,\, Z]$	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[15]
16	$[Y,\, X,\, Z]$	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[16]